Document Management System

DMS | Functional Requirements & Technical Design

DOCUMENT CONTROL	DETAIL
Project	DMS — Document Management System
Deployment	Company intranet; Windows Server / IIS
Technology	ASP.NET Core MVC, C#, Razor, HTML, CSS, JavaScript
Database	Microsoft SQL Server — exactly six application tables
Capacity	~2,000 registered users; concurrency to be assessed
Status	Design baseline; open decisions explicitly marked

1. Purpose and scope

DMS is an intranet-only application for registering controlled PDF documents, extracting first-page metadata, reviewing and approving document revisions, processing plant-user viewing requests, issuing documents, and preserving audit evidence. It is designed for a simple user interface and strict server-side security.

2. Confirmed business rules

- PDF-only uploads. Existing documents generally have a consistent first-page header followed by details; future layouts may vary.
- Extract metadata by searching for words/labels on page one. QA checks and corrects results before saving.
- Detect existing documents by serial number and flag matching document names; a replacement is handled as a revision, not a duplicate master.
- When a new approved revision becomes effective, archive the previous PDF under Obsolete / Plant / Serial Number / Revision. No DMS user, including QA, can see obsolete files in the application.
- Retain minimal obsolete revision metadata and audit records in SQL Server; do not delete archived PDFs.
- Plant users request permission to view a current document. QA approval grants one viewing session; the next visit requires a new request.
- QA must re-enter their own password in a popup before viewing a document or approving a document/access request; this is checked on the server.
- No Plant, Unit, or Department columns in the User Master. Plant and Unit describe documents only.
- No Action column in the database; user interface controls are derived from role and workflow state.

3. Technology and deployment

Layer	Implementation
Web application	ASP.NET Core MVC with Razor (.cshtml)
Frontend	HTML5, CSS3, Bootstrap, vanilla JavaScript
Business logic	C# services and authorization policies
Database	SQL Server with Entity Framework Core
Authentication	Salary-code login with securely hashed password, or company directory if approved
PDF processing	C# PDF text extraction; OCR not in initial MVP
PDF storage	Protected NTFS folders outside IIS wwwroot
Hosting	IIS, internal DNS, HTTPS with company-trusted certificate

All PDF bytes must be delivered by authorized controller endpoints. File paths must never be exposed to users. Network isolation alone is not an authentication or authorization control.

4. Users and authorization

Role	Allowed activities
Plant	Search active document metadata; request access; see own request status; view during an approved session.
QA	Upload and verify PDFs; perform assigned review, approval, viewing, and issuance tasks.
Admin	Manage users and access settings; no automatic document approval authority.

The single AccessRole column supports the simple MVP. If separation of duties is required (uploader cannot approve, distinct reviewer/approver), enforce it with workflow rules and designated role values; a future role-permission model may require additional tables.

5. Six-table SQL Server design

These are six DMS application tables. SQL Server system tables are not included in this count. Columns shown are the proposed core fields; implementation may add technical concurrency and security fields without creating extra application tables.

5.1 DMS_Users

Field	Type / note
UserID	BIGINT IDENTITY PRIMARY KEY
SalaryCode	NVARCHAR(50), UNIQUE, NOT NULL
EmployeeName	NVARCHAR(150), NOT NULL
PasswordHash	NVARCHAR(500), NOT NULL for local login

AccessRole	NVARCHAR(30): Admin / QA / Plant
IsActive	BIT, default 1

PasswordHash must use an approved adaptive password hasher (e.g., ASP.NET Core PasswordHasher). Passwords are never stored in plaintext. Authentication lockout state can be stored as additional columns in this same table.

5.2 DMS_Documents

Field	Type / note
DocumentID	BIGINT IDENTITY PRIMARY KEY
DocumentName	NVARCHAR(255), NOT NULL
SerialNumber	NVARCHAR(100), NOT NULL
Plant	NVARCHAR(50), NOT NULL
DocumentType	NVARCHAR(100)
Unit	NVARCHAR(100)
Category	NVARCHAR(100)
MaterialNumber	NVARCHAR(100), if used
MaterialType	NVARCHAR(100), if used
CurrentRevisionID	BIGINT NULL; current effective revision
CreatedBy / CreatedAt	BIGINT / DATETIME2

Suggested document identity: Plant + SerialNumber, subject to confirmation that serial numbers can repeat between plants. CurrentRevisionID references DMS_Revisions; enforce this relationship carefully to avoid circular insert problems.

5.3 DMS_Revisions

Field	Type / note
RevisionID	BIGINT IDENTITY PRIMARY KEY
DocumentID	BIGINT FOREIGN KEY
RevisionNumber	NVARCHAR(30), NOT NULL
FileName	NVARCHAR(255), original uploaded name
FileLocation	NVARCHAR(1000), protected server path
FileHash	CHAR(64), SHA-256 checksum
Status	Pending / Review / Approved / Active / Obsolete / Rejected
UploadedBy / UploadedAt	BIGINT / DATETIME2
EffectiveAt / ObsoleteAt	DATETIME2 NULL

Unique index: (DocumentID, RevisionNumber). Revision values must not be compared lexicographically unless the organization confirms a numeric scheme.

5.4 DMS_Requests

Field	Type / note
RequestID	BIGINT IDENTITY PRIMARY KEY
DocumentID / RevisionID	BIGINT references
RequestedBy / RequestedAt	BIGINT / DATETIME2
Status	Pending / Approved / Rejected / Viewing / Consumed / Expired
ApprovedBy / ApprovedAt	BIGINT / DATETIME2 NULL
ViewTokenHash	NVARCHAR(128) NULL
ViewStartedAt / ViewExpiresAt	DATETIME2 NULL
Remarks	NVARCHAR(500) NULL

One approved request creates only one viewing session. The server checks request owner, active revision, session token, and expiry for every PDF data request. PDF page/range requests within that same session are permitted.

5.5 DMS_Approvals

Field	Type / note
ApprovalID	BIGINT IDENTITY PRIMARY KEY
DocumentID / RevisionID	BIGINT references
RequestID	BIGINT NULL, for access decisions
WorkflowType	Review / DocumentApproval / AccessApproval / Issue
Decision	Approved / Rejected / Returned / Issued
PerformedBy / PerformedAt	BIGINT / DATETIME2
ReauthenticatedAt	DATETIME2 NULL
Remarks	NVARCHAR(1000) NULL

Each decision creates a separate immutable event. This combined workflow table is a deliberate MVP simplification; issuance-specific details (copy number, quantity, recipient) may need extra columns once the Issue process is confirmed.

5.6 DMS_AuditLogs

Field	Type / note
AuditID	BIGINT IDENTITY PRIMARY KEY
UserID	BIGINT NULL
ActionType	NVARCHAR(100)
DocumentID / RequestID	BIGINT NULL
ActionAt	DATETIME2
IPAddress	NVARCHAR(45)
Details	NVARCHAR(MAX), sanitized structured data

Never record passwords, PDF session tokens, or confidential PDF contents in logs. Use restricted append-only write access and separate backups.

6. PDF import and extraction

Upload → validate extension, content signature and size → save to protected Pending storage → extract text from first page → search configured keywords → parse adjacent/below values → display editable verification form → QA confirms → duplicate/revision check → submit for review.

Field	Example labels to search
Plant	PLANT, PLANT CODE
Serial number	SERIAL NO., DOCUMENT NO., DOC NO.
Document name	DOCUMENT NAME, TITLE
Type	TYPE, DOCUMENT TYPE
Unit	UNIT, UNIT NO.
Category	CATEGORY
Material number	MATERIAL NO., MATERIAL NUMBER
Material type	MATERIAL TYPE
Revision	REVISION, REV. NO.

Because only six tables are allowed, extraction aliases can live in version-controlled application configuration rather than a seventh rules table. Missing or ambiguous fields must be highlighted, never guessed. A redacted sample first page is required to finalize extraction rules.

7. Document lifecycle and obsolescence

New upload → metadata verified → pending review → reviewed → pending document approval → approved → effective/active. A new revision does not replace the current active revision until its effective point. Once effective, the old revision becomes obsolete.

Archive process: stage the old file in Obsolete / Plant / Serial Number / Revision; verify checksum; update revision status and current revision pointer; invalidate grants to the old revision; record audit. Because file moves and SQL updates are not one atomic transaction, use an idempotent, recoverable operation with error handling. Do not expose an old file or mark a new revision active if archival is incomplete.

Archived files are hidden from all DMS screens and application download endpoints. Only specifically authorized IT personnel may access the folder outside DMS. Keep the minimum metadata and audit trail in SQL Server.

8. QA password re-verification

A signed-in QA employee must enter their own account password again when clicking View or Approve. The same rule applies to QA approval of plant viewing requests. The password popup is only a user interface: the backend verifies the password and operation-specific authorization before performing the action.

- Re-authentication is bound to the specific document, revision, action, user, and a short validity window; it cannot authorize a different action.
- For approval, verify the password and commit the decision atomically with the audit record where possible. Reject stale or duplicate submissions.
- For viewing, verify password before issuing a short-lived viewer grant; every PDF request is authorized on the server.
- Rate-limit failed attempts, apply account lockout safeguards, require HTTPS, and never log password values.
- Password re-entry alone is not a certified electronic signature; applicable regulatory controls require separate assessment.

9. Plant request and one-time viewer

Plant employee searches active document metadata → requests access → QA approves after password re-verification → View becomes available → employee opens one authorized PDF viewing session → session expires/ends → request is consumed → next visit requires a new request and QA approval.

A session may include multiple page and HTTP range requests. Refreshing within the same active session need not consume a second approval. Closing a browser tab is not a reliable server signal, so enforce an absolute timeout and/or inactivity timeout. Old links and grants must stop working after expiry or revision obsolescence.

A browser viewer cannot guarantee prevention of screenshots, photography, or client-side capture. Download/print restrictions and exact session timeout remain decisions to confirm.

10. Screens and UI

Screen	Purpose
Login	Salary-code authentication
Dashboard	Role-specific pending work and counts
User Master	Admin management of minimal user accounts
Document Master	Search/filter current document metadata; no database Action field
Upload & Verify	PDF upload and extracted-field confirmation
Review	Review pending document revisions
Approval	Approve/reject document revisions; password popup
Access Requests	Plant requests and QA access decisions
PDF Viewer	One-session authorized PDF view
Issue	Track issuance according to finalized company rules

UI buttons are determined dynamically from the employee role and current workflow state. No unnecessary Action column is stored in SQL Server.

11. Security, operations and performance

- HTTPS throughout the intranet; company-trusted TLS certificate; secure, HttpOnly, SameSite cookies and session timeout.
- Server-side authorization on every endpoint, anti-forgery protection on state-changing forms, parameterized EF Core database access.
- Protected storage outside IIS wwwroot; validate PDF signatures, size, paths and file names; consider malware scanning.
- Database backups, archived-file backups, restore testing, and restricted SQL Server service accounts.
- Pagination, appropriate indexes, asynchronous file delivery and SQL connection pooling for ~2,000 registered employees.
- Measure concurrent usage and PDF traffic before final server sizing; 2,000 accounts does not mean 2,000 simultaneous sessions.

12. MVP acceptance tests

- Only active users can authenticate; disabled users are denied.
- QA can upload a PDF, inspect first-page extracted metadata and correct missing values.
- Duplicate serial/revision combinations are blocked; matching names trigger a warning.
- Document review and approval decisions are recorded with user and timestamp.
- QA View and Approve cannot proceed without successful fresh password verification.
- A plant employee cannot view a PDF without an approved request.
- An approved plant request allows one viewer session; a subsequent visit requires a new approval.
- An obsolete revision is absent from all DMS lists and cannot be retrieved by old URLs.
- Old PDF is preserved in Obsolete / Plant / Serial Number / Revision with matching checksum.
- Audit events and database backups survive ordinary application restarts.

13. Decisions remaining before coding

Topic	Decision required
Login	Salary code + local password or company Active Directory?
PDF fields	Confirm mandatory fields and share redacted first-page sample.
Document identity	Is serial number unique globally or within a plant?
QA duties	Who may review/approve? Must reviewer differ from uploader/approver?
Revision timing	Immediate activation after approval or scheduled effective date?
Plant viewer	Exact session timeout; whether download/print is permitted.
QA viewer	Must QA re-enter password for each distinct View click? Assumed yes.
Issue screen	Does Issue mean electronic release, physical controlled copy, or both?

Infrastructure	Windows/SQL Server versions, storage capacity, concurrent-user estimate.
Compliance	Does GMP / electronic signature validation apply?

14. Delivery plan

Phase 1: MVC project, SQL schema and secure login. Phase 2: document master, upload and extraction. Phase 3: revisions, review and approval. Phase 4: QA step-up authentication and plant access requests. Phase 5: one-time viewer, issue register and obsolete archival. Phase 6: audit, security tests, performance tests and IIS deployment.

Design note: The six-table limit is a project constraint, not a universal database best practice. If requirements grow to complex role assignments, multiple plants per user, electronic signatures, or detailed controlled-copy issuance, revisiting normalization may be necessary.